CODE2DAY · MENTORMIND AI · COMBINED AWS DEPLOYMENT

Building to ₹4.16 per student per month

Implementation architecture and committed cost breakdown for two concurrency tiers on AWS Asia Pacific (Mumbai)

Tier	Registered	Peak concurrent	Budget ceiling	Designed cost	Per student	Result
Tier A	3,000	2,000	₹12,000/mo	**₹11,888/mo**	**₹3.96**	FITS
Tier B	10,000	3,000	₹40,000/mo	**₹39,212/mo**	**₹3.92**	FITS

Designed cost shown on the **pessimistic** path: burst capacity billed at full On-Demand rather than Spot, *and* every registered student assumed to exhaust their full monthly AI allowance. Both tiers also clear the exact 4.16 × N ceiling (₹12,480 and ₹41,600). Students in reality use the platform very unevenly — some an hour, some five, some not at all in a given month — which is modelled in §2.1 and brings the **expected** bills down to ₹9,788 and ₹29,092, or ₹3.26 and ₹2.91 per student.

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Two corrections to every earlier cost document in this series.

1. **The rupee assumption was wrong.** Revisions 1–4 converted at ₹85/USD. The Federal Reserve H.10 release of 4 September 2026 puts the rate at **₹94.49**, with the market near ₹94.80. Every rupee figure in those PDFs therefore understates the bill by about **11.5%**. This document plans at **₹95/USD** and stress-tests to ₹103.
2. **RDS in Mumbai is far more expensive than the public headline rate.** Third-party pricing sites quote `db.m7g.large` at $0.168/hr; that is the us-east-1 rate. The AWS Price List API returns **$0.240/hr** for ap-south-1 — a **43% regional premium**. That single fact is why this design self-manages PostgreSQL on EC2 rather than using RDS.

Verdict. Both tiers fit on the pessimistic Spot-unavailable path. Tier A lands at **₹3.96** per student per month against the ₹4.16 ceiling; Tier B at **₹3.92**. Margins are not symmetric and Tier A is the tight one — see §1.1. The budget survives the rupee weakening to **₹96/USD** (Tier A) and **₹97/USD** (Tier B) before either breaches. Getting there is not a matter of picking cheaper instances — it needs **nine specific changes to the application** (§8), and it gives up **managed database failover, a load balancer, multi-AZ redundancy and most observability** (§10, §11). Those are the honest trade-offs, priced.

1 What the budget is, in absolute money

₹4.16 per registered student per month is a per-head figure; the engineering constraint is the absolute monthly bill it implies. Two ceilings are in play, and this design is built to the tighter one.

Tier	Registered students	Exact ceiling (4.16 × N)	Rounded ceiling (used here)	Rounded ceiling in USD (at ₹95)	Extra slack from rounding
Tier A	3,000	₹12,480	**₹12,000**	$126.32	₹480
Tier B	10,000	₹41,600	**₹40,000**	$421.05	₹1,600

Everything that follows is measured against the **rounded** ceiling — ₹12,000 and ₹40,000 — so there is a further ₹480 / ₹1,600 of unused margin against the arithmetic 4.16 limit.

1.1 Tier A is the harder of the two, and the margins are not symmetric

It is tempting to read Tier B as the demanding case because it has more students and more concurrency. On a per-student budget the opposite is true. A fixed floor of always-on infrastructure — the reserved baseline node, the database, the cache — costs about **$62.23**

per month whether it serves three thousand students or ten thousand. Spread across 3,000 registered students that floor alone is ₹1.97 per student; across 10,000 it is ₹0.59. Tier A therefore starts roughly ₹1.38 per student behind before a single inference call is made.

Margin against…	Tier A, Spot	Tier A, On-Demand	Tier B, Spot	Tier B, On-Demand
the rounded ceiling (₹12,000 / ₹40,000)	₹646 5.4% of the ceiling	₹112 0.9% of the ceiling	₹1,855 4.6% of the ceiling	₹788 2.0% of the ceiling
the exact 4.16 ceiling (₹12,480 / ₹41,600)	₹1,126 9.0% of the ceiling	₹592 4.7% of the ceiling	₹3,455 8.3% of the ceiling	₹2,388 5.7% of the ceiling

Read this margin honestly. On the On-Demand fallback path Tier A has only **₹112** of room against the rounded ₹12,000 — 0.9% — though it keeps **₹592** against the arithmetic ₹12,480 ceiling. That is thin enough that Tier A should be treated as a budget that needs active management: the graded-evaluation quota is the dial, and each evaluation removed from the monthly allowance recovers about ₹509. Tier B, by contrast, has genuine slack at ₹788. Note that both of those margins are the **worst case**, in which every registered student exhausts their allowance; §2.1 shows the expected margins under realistic uneven usage, which are far wider (₹2,212 at Tier A).

2 Load model

Cost is driven by two independent numbers, and conflating them is the most common way these estimates go wrong. **Peak concurrency** sets how much compute must exist at the busiest moment — it buys capacity in discrete steps. **Daily active users** sets how much inference gets consumed — it scales linearly and continuously. A tier can be cheap on one and expensive on the other.

Parameter	Tier A	Tier B	Why it is set this way
Registered students	3,000	10,000	The billing base. Quotas are priced against this, not against DAU, so the ceiling holds even in the worst case where every registered student consumes their full monthly allowance.
Peak concurrent users	2,000	3,000	Stated as fluctuating. Sizing targets the peak, not the average, and the design absorbs swings via a scheduled burst tier plus free target-tracking autoscaling.
Daily active users	2,850	7,000	Tier A: 2,000 of 3,000 online at once implies essentially the whole cohort uses it daily (95%). Tier B: 3,000 of 10,000 concurrent implies a staggered timetable (70%).
Active window	8 h/day	8 h/day	The single largest cost lever. 8 h × 22 weekdays = 176 of 730 hours, or 24% of the month. Only the burst tier is paid for during those 176 hours.
Requests per engaged user	1 per 25 s	1 per 25 s	Page views, autosaves and polling from a student who is reading and typing, not idling.
Required peak request rate	80 req/s	120 req/s	Peak concurrency divided by think time. This, not the raw session count, is what the web tier has to serve.
Code submissions per active student per day	6	6	Code2Day judge executions. A quarter are repeats of identical code and are served from a content-hash cache.

Why the 8-hour window matters so much. A design that runs its full peak fleet 24×7 pays for 730 hours. A design that reserves a modest always-on baseline and adds burst capacity only during the teaching day pays for 730 hours of the baseline plus 176 hours of the burst. On Tier B that is the difference between four nodes billed continuously and two nodes billed continuously plus two billed for 24% of the month.

2.1 Students will not all use it the same amount, and that is priced in

Some students will spend an hour on the platform, some will spend five, and some will not open it at all in a given month. No real cohort uses a learning platform uniformly, and a cost model that assumes it does will be wrong in both directions at once — under-provisioned for the heavy tail and over-charged for the dormant one. This spread is accounted for in three distinct places in this design, and it is worth being explicit about each.

How the spread is handled	Mechanism
Compute is sized on the peak, never the average	The web tier is built for the busiest simultaneous moment — 2,000 and 3,000 concurrent — and then carries 3.0× and 4.0× headroom on top of that. The heavy five-hour-a-day students are exactly who that capacity exists for. A student who uses one hour costs the same nothing extra, because the capacity is already reserved.
Inference is charged as if *every* student maxed out	The AI lines in §5 and §6 are priced at **registered students × the full monthly quota** — 3,000 × 6 graded evaluations at Tier A, 10,000 × 8 at Tier B. That is a deliberate worst case: it assumes all 3,000 and all 10,000 students exhaust their entire allowance every month, which no cohort does. Because the quota is a hard server-side cap, a heavy user cannot exceed it, and every light or dormant user comes in under it. The AI figure is therefore a **ceiling**, not a forecast.
Non-users are already excluded from the daily load	Daily active users are set below registered users — 2,850 of 3,000 at Tier A and 7,000 of 10,000 at Tier B — so the judge executions, transcript writes and egress are modelled on the students who actually turn up, not on the roll.

The usage mix this was tested against

Cohort	Share	Hours per day	Active days per month	Quota consumed	Behaviour
Dormant	10%	none	0	0%	Registered but does not log in at all this month.
Light	30%	~1 h	8	25%	Dips in for about an hour, a couple of times a week.
Moderate	40%	~3 h	16	65%	Regular use on most teaching days.
Heavy	20%	~5 h	22	100%	In every day, exhausts the monthly quota.
Weighted average	**100%**	**—**	**—**	**53.5%**	**About 44 student-hours per registered student per month.**

What that mix does to the bill

Splitting each tier into the fixed part — reserved compute, database, cache, which cost the same whether students show up or not — and the variable part — inference, judge executions, request and egress charges, which only accrue on use — gives the expected monthly bill under this mix alongside the worst case.

	Tier A	Tier B
Fixed, regardless of how much anyone uses it	$77.61	$183.68
Variable, if every student exhausts their quota	$47.53	$229.08
Variable, expected under the mix above (53.5% of quota)	$25.43	$122.56
Worst case — every student maxes out	**₹11,888**	**₹39,212**
Expected bill under the mix above	**₹9,788**	**₹29,092**
Expected per registered student per month	₹3.26	₹2.91
Expected bill as a share of the worst case	82%	74%
Expected headroom against the ceiling	₹2,212	₹10,908
Expected monthly saving versus the worst case	₹2,100	₹10,120

Every headline figure in this document is the worst case. ₹11,888 at Tier A and ₹39,212 at Tier B assume all 3,000 and all 10,000 registered students consume their entire monthly allowance *and* that Spot capacity is unavailable. Under the realistic mix above — a dormant tenth, a light third, a heavy fifth — the expected bills are **₹9,788** and **₹29,092**, or **₹3.26** and **₹2.91** per student. That is where the real safety margin lives: Tier A's uncomfortably thin ₹112 of worst-case headroom becomes about **₹2,212** in expectation.

The one thing this mix does **not** soften is peak concurrency. If the heavy cohort all arrive in the same hour — a scheduled lab, an assessment window, a deadline — the fleet still has to serve 2,000 and 3,000 simultaneous sessions, which is why the compute is sized on the peak and not on these averages. The mix reduces the bill; it does not reduce the capacity requirement.

3 Implementation architecture

One shape serves both tiers; only the sizing changes. Every box is either reserved for three years, billed by the second, or free.

```
                     STUDENT BROWSERS  (2,000 / 3,000 concurrent at peak)
                                   |
                                   |  HTTPS
                   +---------------v-----------------+
                   |    CLOUDFLARE  (Free plan)      |   Rs 0
                   |  DNS - TLS - CDN - WAF          |   . static bundle cached at edge
                   |  rate limiting - DDoS           |   . unmetered egress to users
                   |  Web Speech API used in-browser |   . per-IP + per-user rate limits
                   +---------------+-----------------+
                                   |  Cloudflare Tunnel (cloudflared)
                                   |  no ALB - no public IPv4 - no NAT Gateway
   +-------------------------------v-----------------------------------+
   |                    APP TIER  (private subnet)                     |
   |                                                                   |
   |   BASELINE  24x7, 3yr Reserved        BURST  176 h/mo, Spot       |
   |   Tier A: 1 x c7g.xlarge  (4 vCPU)    Tier A: 1 x c7g.large       |
   |   Tier B: 2 x c7g.xlarge  (8 vCPU)    Tier B: 2 x c7g.large       |
   |                                                                   |
   |   gunicorn + gevent (non-blocking on AI waits)                    |
   |   Celery workers, same nodes - AI calls are I/O-bound, near-zero  |
   |   CPU while awaiting Groq / Bedrock                               |
   +--+---------------+-----------------+----------------+-------------+
      |               |                 |                |
+-----v-----+  +------v------+  +-------v-------+  +-----v----------------+
| POSTGRES  |  | ELASTICACHE |  |    LAMBDA     |  |  EXTERNAL  INFERENCE |
| on EC2    |  |   VALKEY    |  |   (arm64)     |  |                      |
| self-mgd  |  |             |  |               |  |  Groq whisper-       |
|           |  | sessions    |  | Code2Day      |  |  large-v3-turbo      |
| A: m7g.   |  | hot reads   |  | judge, in a   |  |  $0.04/audio-hour    |
|    large  |  | Celery      |  | fresh sandbox |  |                      |
| B: m7g.   |  |  broker     |  | per submit    |  |  Groq gpt-oss-20b    |
|    xlarge |  | judge       |  |               |  |  practice + agents   |
|  + hot    |  |  result     |  | scales to 0   |  |                      |
|  standby  |  |  cache      |  | 400k GB-s free|  |  DeepSeek V3.2 via   |
+-----+-----+  +-------------+  +---------------+  |  Bedrock Mantle,     |
      |                                            |  graded path only    |
      |  pgBackRest: base backup + WAL             +----------------------+
      v
+------------------+
|   S3             |   Standard-IA for WAL archive - PITR to any point in 14 days
|                  |   Standard for the static bundle and stored transcripts
+------------------+                                    audio is never persisted
```

3.1 Why each choice is what it is

Layer	Choice	Reason
Edge	Cloudflare Free, not CloudFront or ALB	Gives TLS, CDN, WAF, DNS and per-user rate limiting at no cost, and its egress to users is unmetered. Only API JSON then leaves AWS, which at Tier A stays inside the 100 GB/month free allowance entirely. Rate limiting at the edge is also the cheapest possible defence for the graded-evaluation endpoint, whose cost is otherwise attacker-controlled.
Ingress	Cloudflare Tunnel, not an Application Load Balancer	Saves the $17.45/month ALB base charge plus LCU, the $3.60/month public IPv4 address and any NAT Gateway. App nodes need no inbound security-group rule at all. The cost is a hard dependency on Cloudflare and the loss of AWS-native health-check-driven target replacement.
Compute family	Graviton c7g, not x86	c7g.xlarge gives 4 vCPU at $0.0445/hr on a 3-year No-Upfront reservation. Requires dropping the x86-only torch wheel — see §8, lever 10.
Commitment	3-year No-Upfront Standard Reserved Instances	Worth 55–58% off On-Demand with **zero cash up front**. This is the largest single infrastructure saving in the design, and it is also the biggest strategic commitment: without it Tier A costs roughly 45% more and breaches the ceiling.
Burst capacity	Spot for the 176-hour teaching window	Stateless web nodes tolerate a 2-minute interruption notice. Both tiers are costed on the assumption that Spot is **unavailable** and the burst falls back to On-Demand, so the budget never depends on Spot actually being there.
Database	Self-managed PostgreSQL on EC2, not RDS	RDS db.m7g.large in Mumbai is $0.240/hr On-Demand. The same 2 vCPU / 8 GiB as raw EC2 m7g.large reserved is $0.0265/hr — a 4.1× difference on the single largest infrastructure line. This is the choice that makes ₹4.16 arithmetically possible, and it transfers patching, backup, failover and monitoring to your team.
Durability	pgBackRest base backup plus WAL archive to S3 Standard-IA	Point-in-time recovery to any moment in the last 14 days for about $1/month. This replaces RDS automated backups; it does not replace automatic failover.
Cache	ElastiCache Valkey, not Redis, not self-hosted	Valkey is priced 20% below Redis for identical nodes and, unlike RDS, offers a 3-year No-Upfront reservation. Kept managed because it holds sessions: losing it logs everyone out, and it is only $5.84–$12.26/month.
Code execution	Lambda arm64, not EC2 judge workers	Marginally cheaper than dedicated capacity at these volumes, but chosen for isolation: every submission of untrusted student code runs in a fresh microVM. arm64 is 20% cheaper than x86 and the 400,000 GB-s monthly free tier covers most of Tier A.
Async model	gunicorn + gevent, with Celery for the graded path	The existing 12 synchronous workers hold a whole worker for the entire Whisper-plus-LLM round trip. Under gevent an in-flight AI call costs a greenlet, not a worker, which is what lets 6 vCPU absorb 2,000 concurrent sessions.

4 Does this much compute actually serve that many users?

A budget that fits but falls over is worthless. The check below converts concurrency into a request rate and compares it against the fleet. The stated tolerance for higher latency is what makes the margins below acceptable rather than reckless.

Check	Tier A	Tier B
Peak concurrent sessions	2,000	3,000
Required request rate at peak	80 req/s	120 req/s
vCPU available at peak (baseline + burst)	6	12
Serving capacity at peak (~40 req/s per vCPU)	~240 req/s	~480 req/s
Headroom at peak	3.0×	4.0×
vCPU on the always-on baseline alone	4	8
Headroom if the burst tier never launches	2.0×	2.7×
Concurrency at which the peak fleet saturates	~6,000	~12,000

> **The fluctuation is covered.** Both tiers hold 3.0× and 4.0× headroom at their stated peak, and saturate only around 6,000 and 12,000 concurrent sessions. Even with the burst tier entirely absent — a total Spot outage, or an autoscaling failure — the reserved baseline alone still carries 2.0× and 2.7× the required rate. Concurrency rising or falling by half, as you expect it to, changes nothing structural.

The 40 req/s per vCPU figure assumes sessions and hot reads are served from Valkey rather than PostgreSQL, and that the N+1 query patterns are addressed. It is a mixed-workload figure for cached JSON plus ORM reads on Graviton, not a benchmark of this specific codebase — it must be confirmed by the load test in §9, phase 5. If the real figure is 20 req/s per vCPU, Tier A headroom falls to 1.5×, which still holds.

5 Tier A cost breakdown — 3,000 registered, 2,000 peak concurrent

Priced on the **On-Demand burst fallback**, the pessimistic path. Every AWS rate is from the AWS Price List API for ap-south-1 and is listed in §12.

Component	Sizing and rate basis	USD / month
COMPUTE		**$45.02**
App tier baseline - 1 x c7g.xlarge	4 vCPU / 8 GiB, 24x7, 3yr No-Upfront RI @ $0.0445/hr x 730 h	$32.48
App tier burst - 1 x c7g.large	+2 vCPU for 176 h/mo (On-Demand) @ $0.0491/hr	$8.64
App node root volumes - gp3	1 x 30 GB always-on + 1 x 30 GB pro-rated 176/730 h = 37.2 GB	$3.40
Golden AMI snapshot	~10 GB x $0.05/GB-mo	$0.50
DATABASE		**$23.91**
PostgreSQL primary - 1 x m7g.large	2 vCPU / 8 GiB, self-managed, 24x7, 3yr No-Upfront RI @ $0.0265/hr	$19.34
Primary data volume - 50 GB gp3	50 GB x $0.0912/GB-mo (3,000 IOPS + 125 MB/s included)	$4.56
CACHE / QUEUE		**$5.84**
ElastiCache Valkey - cache.t4g.micro	0.5 GiB, sessions + hot-read cache + Celery broker, 3yr No-Upfront @ $0.008/hr	$5.84
CODE EXECUTION		**$0.31**
Lambda (arm64) - Code2Day judge	376,200 submits - 25% Valkey content-hash cache hits = 282,150 cold runs x 1,024 MB x 1.5 s = 423,225 GB-s; 400k GB-s free tier applied	$0.31
STORAGE		**$0.93**
S3 Standard - static bundle + transcripts	5 GB x $0.025/GB-mo	$0.12
S3 Standard-IA - pgBackRest base + WAL archive	25 GB x $0.0138/GB-mo (PITR to any point in 14 days)	$0.34
S3 requests	62,700 PUT (1 transcript/student/day; judge verdicts go inline to Postgres, not S3) + ~376,200 GET	$0.46
NETWORK		**$0.00**
Data transfer out to internet (API JSON only)	61 GB/mo - 100 GB free = 0 GB x $0.1093/GB	—
Cloudflare Free - CDN, TLS, WAF, DNS, unmetered egress to users	static assets + rate limiting served at the edge; replaces ALB + CloudFront	—
Cloudflare Tunnel - ingress to app nodes	cloudflared sidecar on each node; no ALB, no public IPv4, no NAT Gateway	—
OPS		**$2.37**
CloudWatch Logs (Infrequent Access class)	2 GB ingest x $0.335/GB + 10 GB stored x $0.03/GB-mo	$0.97
CloudWatch alarms	6 standard alarms x $0.1	$0.60
Secrets Manager	2 secrets x $0.4/mo	$0.80
AI		**$46.76**
Graded speaking evaluation (Whisper turbo + DeepSeek V3.2)	6/student/mo x 3,000 students = 18,000 evals x $0.00179	$32.13
Practice + chat turns (browser ASR + gpt-oss-20b)	25/student/mo x 3,000 = 75,000 turns x $0.000126	$9.45
Resume / interview / learning-path / advisor (gpt-oss-20b)	5/student/mo x 3,000 = 15,000 calls x $0.000345	$5.17
Model router / task classifier	keyword-based, no LLM call (load_balancer_agent.py:47)	—
INFRASTRUCTURE SUBTOTAL		**$78.38**
AI / INFERENCE SUBTOTAL		**$46.76**
TOTAL PER MONTH — On-Demand burst fallback		**$125.14**

Component	Sizing and rate basis	USD / month
TOTAL PER MONTH IN RUPEES (at ₹95/USD)		**₹11,888**
PER REGISTERED STUDENT PER MONTH		**₹3.96**

IF SPOT CAPACITY IS AVAILABLE

₹11,354

per month — ₹3.78 per student — ₹136,254/year

BUDGET POSITION (ON-DEMAND PATH)

₹112 under

₹11,888 spent of the ₹12,000 ceiling — 0.9% margin

Summary	Spot burst	On-Demand burst
Infrastructure	$72.76	$78.38
AI / inference	$46.76	$46.76
Total, USD	$119.52	$125.14
Total, rupees	₹11,354	₹11,888
Per registered student per month	₹3.78	₹3.96
Per registered student per year	₹45.42	₹47.55
Annual run rate	₹136,254	₹142,657
Headroom against the ₹12,000 ceiling	₹646	₹112
Rupee rate at which the ceiling breaks	₹100.40/USD	₹95.89/USD

What each student gets for ₹3.96 a month. **6** fully graded speaking evaluations (Groq Whisper transcription plus a DeepSeek V3.2 assessment), **25** practice or conversation turns using in-browser speech recognition with a gpt-oss-20b reply, and **5** resume, interview, learning-path or advisor calls. Beyond the graded quota the feature stays open as unscored self-practice with browser transcription. These quotas are the budget: they are enforced server-side, and they are the only reason the AI line is a ceiling rather than an estimate.

6 Tier B cost breakdown — 10,000 registered, 3,000 peak concurrent

Priced on the **On-Demand burst fallback**, the pessimistic path. Every AWS rate is from the AWS Price List API for ap-south-1 and is listed in §12.

Component	Sizing and rate basis	USD / month
COMPUTE		**$89.54**
App tier baseline - 2 x c7g.xlarge	8 vCPU / 16 GiB, 24x7, 3yr No-Upfront RI @ $0.0445/hr x 730 h	$64.97
App tier burst - 2 x c7g.large	+4 vCPU for 176 h/mo (On-Demand) @ $0.0491/hr	$17.28
App node root volumes - gp3	2 x 30 GB always-on + 2 x 30 GB pro-rated 176/730 h = 74.5 GB	$6.79
Golden AMI snapshot	~10 GB x $0.05/GB-mo	$0.50
DATABASE		**$76.28**
PostgreSQL primary - 1 x m7g.xlarge	4 vCPU / 16 GiB, self-managed, 24x7, 3yr No-Upfront RI @ $0.053/hr	$38.69
Primary data volume - 100 GB gp3	100 GB x $0.0912/GB-mo (3,000 IOPS + 125 MB/s included)	$9.12
Streaming standby - 1 x m7g.large	2 vCPU / 8 GiB, hot standby for failover, 3yr No-Upfront RI	$19.34
Standby data volume - 100 GB gp3	100 GB x $0.0912/GB-mo	$9.12
CACHE / QUEUE		**$12.26**
ElastiCache Valkey - cache.t4g.small	1.37 GiB, sessions + hot-read cache + Celery broker, 3yr No-Upfront @ $0.0168/hr	$12.26
CODE EXECUTION		**$8.53**
Lambda (arm64) - Code2Day judge	924,000 submits - 25% Valkey content-hash cache hits = 693,000 cold runs x 1,024 MB x 1.5 s = 1,039,500 GB-s; 400k GB-s free tier applied	$8.53
STORAGE		**$2.59**
S3 Standard - static bundle + transcripts	14 GB x $0.025/GB-mo	$0.35
S3 Standard-IA - pgBackRest base + WAL archive	80 GB x $0.0138/GB-mo (PITR to any point in 14 days)	$1.10
S3 requests	154,000 PUT (1 transcript/student/day; judge verdicts go inline to Postgres, not S3) + ~924,000 GET	$1.14
NETWORK		**$5.51**
Data transfer out to internet (API JSON only)	150 GB/mo - 100 GB free = 50 GB x $0.1093/GB	$5.51
Cloudflare Free - CDN, TLS, WAF, DNS, unmetered egress to users	static assets + rate limiting served at the edge; replaces ALB + CloudFront	—
Cloudflare Tunnel - ingress to app nodes	cloudflared sidecar on each node; no ALB, no public IPv4, no NAT Gateway	—
OPS		**$4.14**
CloudWatch Logs (Infrequent Access class)	4 GB ingest x $0.335/GB + 20 GB stored x $0.03/GB-mo	$1.94
CloudWatch alarms	10 standard alarms x $0.1	$1.00
Secrets Manager	3 secrets x $0.4/mo	$1.20
AI		**$213.91**
Graded speaking evaluation (Whisper turbo + DeepSeek V3.2)	8/student/mo x 10,000 students = 80,000 evals x $0.00179	$142.81
Practice + chat turns (browser ASR + gpt-oss-20b)	40/student/mo x 10,000 = 400,000 turns x $0.000126	$50.40
Resume / interview / learning-path / advisor (gpt-oss-20b)	6/student/mo x 10,000 = 60,000 calls x $0.000345	$20.70
Model router / task classifier	keyword-based, no LLM call (load_balancer_agent.py:47)	—
INFRASTRUCTURE SUBTOTAL		**$198.85**

Component	Sizing and rate basis	USD / month
AI / INFERENCE SUBTOTAL		**$213.91**
TOTAL PER MONTH — On-Demand burst fallback		**$412.76**
TOTAL PER MONTH IN RUPEES (at ₹95/USD)		**₹39,212**
PER REGISTERED STUDENT PER MONTH		**₹3.92**

IF SPOT CAPACITY IS AVAILABLE

₹38,145

per month — ₹3.81 per student — ₹457,739/year

BUDGET POSITION (ON-DEMAND PATH)

₹788 under

₹39,212 spent of the ₹40,000 ceiling — 2.0% margin

Summary	Spot burst	On-Demand burst
Infrastructure	$187.62	$198.85
AI / inference	$213.91	$213.91
Total, USD	$401.53	$412.76
Total, rupees	₹38,145	₹39,212
Per registered student per month	₹3.81	₹3.92
Per registered student per year	₹45.77	₹47.05
Annual run rate	₹457,739	₹470,546
Headroom against the ₹40,000 ceiling	₹1,855	₹788
Rupee rate at which the ceiling breaks	₹99.62/USD	₹96.91/USD

What each student gets for ₹3.92 a month. **8** fully graded speaking evaluations (Groq Whisper transcription plus a DeepSeek V3.2 assessment), **40** practice or conversation turns using in-browser speech recognition with a gpt-oss-20b reply, and **6** resume, interview, learning-path or advisor calls. Beyond the graded quota the feature stays open as unscored self-practice with browser transcription. These quotas are the budget: they are enforced server-side, and they are the only reason the AI line is a ceiling rather than an estimate.

7 Where the inference money actually goes

The AI line is 37% of Tier A and 52% of Tier B, so it deserves to be understood per unit rather than per month. All three figures below are for one complete user-visible action.

Action	Composition	Cost each	In paise
Graded speaking evaluation	45 s of audio through whisper-large-v3-turbo, then one DeepSeek V3.2 call with a prompt-cached rubric	\$0.00179	16.958 p
• transcription component	45 s at \$0.04 per audio-hour	\$0.00050	4.750 p
• LLM input component	1,900 tokens in, of which 1,300 are a cache hit charged at 10% of base	\$0.00045	4.300 p
• LLM output component	450 tokens of trimmed JSON, billed at 2.98× the input rate	\$0.00083	7.909 p
Practice / conversation turn	browser Web Speech API for transcription (free), one gpt-oss-20b call	\$0.00013	1.197 p
Resume / interview / learning-path / advisor call	one gpt-oss-20b call, 2,200 in and 600 out	\$0.00034	3.277 p

> **The output tokens are the whole game.** On a graded evaluation the 450 output tokens cost \$0.00083 while the entire 1,900-token input costs \$0.00045. Output is priced at 2.98× input, and DeepSeek reasoning mode multiplies output volume by three to ten times *without changing the API surface*. Leaving reasoning enabled by accident would take the Tier B AI line from \$213.91 to somewhere between \$470.60 and \$1,283.45 per month and destroy the budget on its own. It must be explicitly disabled and asserted in a test.

Against the code as it stands today — three sequential Committee-of-Experts calls, no prompt caching, verbose JSON — one graded evaluation costs **\$0.00903**. The tuned path costs **\$0.00179**. That **5.1×** reduction is what §8 buys, and without it neither tier fits.

8 The changes the budget depends on

These are not optimisations to consider later. The costs in §5 and §6 assume they are all in place. Shipping the current code onto this infrastructure would exceed both ceilings.

#	Change	Where	Effect
1	**Collapse Committee-of-Experts 3 LLM calls to 1**	communication_agent.py:125-207 (coe_evaluate_text), 297-370 (chat_practice)	-67% of LLM tokens on every graded evaluation
2	**Prompt-cache the rubric / system prompt**	1,300 of 1,900 input tokens read at 10% of base rate	-34% of remaining input cost
3	**Trim the response JSON schema**	23 response_format={"type":"json_object"} call sites; output bills at 2.98x input	900 -> 450 output tokens, -50% of the dominant term
4	**Hard-disable DeepSeek reasoning mode**	reasoning tokens bill as output at 3-10x volume, silently	prevents a 3-10x blow-out of the largest AI line
5	**Browser Web Speech API for practice, Groq Whisper only when graded**	ConversationPartner.jsx:160 already does browser ASR	ASR spend scales with the graded quota, not with total practice
6	**Route non-graded work to Groq gpt-oss-20b, DeepSeek only for grading**	groq_client.py:100 provider= branch already exists	gpt-oss-20b is 8.3x cheaper on input and 6.2x cheaper on output
7	**Keyword task router instead of an LLM classifier**	load_balancer_agent.py:44 asks for a "classifier" model key that does not exist; :47 fallback already works	removes one LLM call per routed request
8	**Server-side per-student quota on the graded endpoint**	the speaking endpoint is currently unthrottled	turns the AI bill from attacker-controlled into a hard ceiling
9	**Valkey content-hash cache on judge submissions**	students re-run identical code	-25% of Lambda executions
10	**Remove the x86-only torch dependency**	`torch==2.10.0+cpu` has no aarch64 wheel; `semantic_matcher.py:24` lazy-loads MiniLM and `:36` already falls back to Jaccard similarity	Unblocks Graviton entirely. Without this the whole design reverts to x86 and loses roughly 10–20% of compute cost.
11	**Move sessions and hot reads to Valkey; set `CONN_MAX_AGE`; switch to gevent workers**	Sessions are in PostgreSQL today; there is no `CONN_MAX_AGE`, and 12 synchronous workers each block for a full AI round trip	The difference between 2 vCPU of database coping and collapsing, and between 6 vCPU absorbing 2,000 concurrent sessions and queueing them.
12	**Delete the fabricated-transcript fallback**	`communication_agent.py:377` `_fallback_transcript()` returns a fixed sentence when ASR fails, and the pipeline then scores it as though it were the student	Not a cost item. It silently produces fake grades and must not reach production at any budget.

Two findings that survive every revision of this analysis. First, there is no read routing anywhere in either codebase — no `DATABASE_ROUTERS`, no `.using()` — so the Tier B standby earns nothing as a read replica until that is implemented; it is bought purely for failover. Second, `CORS_ALLOW_ALL_ORIGINS = True` at `settings.py:154` overrides the allowlist above it while `CORS_ALLOW_CREDENTIALS` is also true. That combination must be fixed before this is exposed to 10,000 students.

9 Implementation order

Sequenced so that each phase is independently verifiable and the expensive commitments come last, after the load test has confirmed the sizing.

Phase	Work	Detail	Gate before moving on
1	**Make the AI spend bounded**	Server-side per-student quota on the graded endpoint; Cloudflare rate limits per IP and per user; hard-disable DeepSeek reasoning mode; log input, output and cached token counts on every call.	A synthetic abuse run cannot push spend past the quota.
2	**Cut the cost per evaluation**	Collapse the Committee-of-Experts to one call; enable prompt caching on the rubric; trim the JSON schemas at the 23 response_format sites; route practice and agent traffic to gpt-oss-20b and reserve DeepSeek V3.2 for grading; keyword task router.	Measured cost per graded evaluation is at or below $0.00179.
3	**Make the app horizontally scalable**	Sessions and hot reads into Valkey; set CONN_MAX_AGE; gunicorn to gevent workers; Celery for the graded path so no HTTP worker blocks on inference; delete _fallback_transcript(); fix CORS.	Two app nodes behind one tunnel serve traffic with sessions surviving a node kill.
4	**Port to Graviton**	Remove torch and sentence-transformers; confirm the Jaccard path is acceptable or replace the embedding step with a hosted call; rebuild the image for arm64; move the judge to Lambda arm64 with a Valkey content-hash result cache.	Full test suite green on aarch64; judge isolation verified.
5	**Load test against the real sizing**	k6 or Locust at 2,000 and 3,000 concurrent against the reserved baseline alone, then with the burst tier. Confirm the 40 req/s per vCPU assumption.	Baseline-only headroom is at least 1.5×. This is the gate for phase 6.
6	**Commit the money**	Only now buy the 3-year No-Upfront Reserved Instances for the baseline and the Valkey node, sized by what phase 5 measured. Configure scheduled scaling for the 8-hour window plus free target-tracking for fluctuation.	Reservation coverage report shows near-100% baseline coverage.
7	**Make the database survivable**	pgBackRest base backup plus WAL archive to S3; a restore rehearsal with a stopwatch; streaming standby at Tier B; alarms on replication lag, disk, and the AI spend counter.	A timed restore from S3 has actually been performed and its RTO written down.

Do not buy the reservations first. A 3-year No-Upfront Standard RI is a 36-month payment obligation. It is the largest saving in this design and the least reversible decision in it. Phase 5 exists to make sure you reserve the right size.

10 What the budget did not buy

Each row is a real alternative, priced at the verified Mumbai rate, that was rejected because it does not fit. This is the honest cost of ₹4.16.

Area	What you get	Cost	What you gave up	Would have cost	Saved
Relational database	Self-managed PostgreSQL on m7g.large EC2 (3yr No-Upfront RI)	$19.34	RDS PostgreSQL db.m7g.large Single-AZ (1yr No-Upfront RI)	$134.61	**$115.27**
Database high availability	None at Tier A - pgBackRest PITR restore, RTO 30-60 min	—	RDS Multi-AZ db.m7g.large (1yr No-Upfront RI), automatic failover	$134.61	**$134.61**
Ingress / load balancing	Cloudflare Tunnel + Cloudflare Free (cloudflared per node)	—	Application Load Balancer, 2 LCU average	$29.13	**$29.13**
Database storage	EBS gp3 on EC2	$9.12	RDS gp3 Single-AZ managed storage	$13.10	**$3.98**
Egress path	Cloudflare Free (unmetered to users); only API JSON leaves AWS	—	Direct AWS egress for all 1,232 GB/mo at Tier B	$134.66	**$134.66**
Observability	4 GB/mo logs in the Infrequent Access class, 10 alarms, no dashboards	$1.34	Full-fidelity logs at 60 GB/mo Standard class + APM	$40.20	**$38.86**
TOTAL MONTHLY COST AVOIDED BY THESE SIX CHOICES					**$456.50**

The consequences, stated plainly: You take over patching, failover, PITR and monitoring. RDS in Mumbai costs 4.1x the raw EC2. ▪ An AZ or instance loss at Tier A is a manual restore, not a 60-second failover. ▪ Hard dependency on Cloudflare; no AWS-native health-check target replacement. ▪ RDS storage is 1.44x the EBS rate for the same 100 GB. ▪ Static assets must actually be cacheable at the edge; a cache-busting bug re-prices this. ▪ Mumbai CloudWatch Logs Standard ingest is $0.67/GB. No tracing, no APM, no dashboards.

11 What runs, what is degraded, what is off

Everything is deployed. Nothing in either product is removed. What changes is the generosity of the AI features and the thickness of the operational safety net.

Capability	Status	At ₹4.16 per student per month
Code2Day: problems, editor, submissions	Full	Full function. Judge runs in an isolated Lambda microVM per submission; identical resubmissions are served from a content-hash cache.
Code2Day: leaderboards, contests, dashboards	Full	Full function, served through the Valkey hot-read cache.
Auth, JWT, token blacklist	Full	Full function. Sessions move from PostgreSQL to Valkey.
MentorMind: graded speaking evaluation	Reduced	Quota-limited to 6/month (Tier A) and 8/month (Tier B). Scored by a single DeepSeek V3.2 call instead of a three-expert committee, so scores are less robust to model idiosyncrasy. Audio capped at 45 seconds.
MentorMind: conversation partner and practice	Reduced	Unlimited as self-practice, but transcription is the browser Web Speech API rather than Whisper — lower accuracy, Chromium and Edge only, no word timings — and replies come from gpt-oss-20b rather than DeepSeek.
MentorMind: resume, interview, learning path, advisor	Reduced	5/6 calls per student per month on gpt-oss-20b. Noticeably weaker than DeepSeek on long reasoning, and the resume semantic matcher falls back to Jaccard similarity once torch is removed.
Audio recordings retained for replay or appeal	Not bought	Not retained. Audio is transcribed and discarded; only the transcript and score are stored. This also sidesteps the 30-day retention that Bedrock Mantle applies by default when `store` is left true.
RAG semantic search	Reduced	The current implementation pulls every chunk for a source type into Python and scores it brute-force. It works at these data volumes but is a latency cliff as content grows. pgvector is the fix and it is free — just not scheduled here.
Database automatic failover	Not bought	Tier A has none: recovery is a pgBackRest restore, RTO 30–60 minutes. Tier B has a streaming standby, but promotion is manual.
Multi-AZ redundancy	Not bought	Single availability zone at both tiers. An AZ failure is a full outage.
Read replicas serving read traffic	Not bought	Pointless until read routing exists in the code. The Tier B standby is for failover only.
Load balancer	Not bought	No ALB. Cloudflare Tunnel provides ingress and health-based routing across tunnel replicas. A Cloudflare outage is a full outage.
Observability: APM, tracing, dashboards	Not bought	None. 2 GB (Tier A) and 4 GB (Tier B) of logs per month in the Infrequent Access class, plus a handful of alarms. Mumbai charges $0.67/GB for Standard log ingest, which is why.
Security services: WAF beyond Cloudflare, GuardDuty, Config, Shield Advanced	Not bought	None. Cloudflare Free provides basic WAF rules, DDoS absorption and rate limiting.
Staging environment on AWS	Not bought	Not budgeted. Deploy from CI to production, or accept a second bill.

12 Risk register and currency sensitivity

12.1 Rupee sensitivity

The bill is incurred in dollars and the budget is set in rupees, so the exchange rate is a live risk rather than a footnote. The table shows the monthly total at each rate against the rounded ceilings.

₹/USD	Tier A, Spot	Tier A, On-Demand	Tier B, Spot	Tier B, On-Demand	Status
94.49 Fed H.10, 4 Sep 2026	₹11,294 ₹3.76/student	₹11,824 ₹3.94/student	₹37,940 ₹3.79/student	₹39,002 ₹3.90/student	all four within the rounded ceilings
95.00 planning rate	₹11,354 ₹3.78/student	₹11,888 ₹3.96/student	₹38,145 ₹3.81/student	₹39,212 ₹3.92/student	all four within the rounded ceilings
98.00 stress	₹11,713 ₹3.90/student	₹12,263 ₹4.09/student	₹39,349 ₹3.93/student	₹40,450 ₹4.05/student	rounded ceilings breached; still inside the exact 4.16 limit
100.00 stress	₹11,952 ₹3.98/student	₹12,514 ₹4.17/student	₹40,153 ₹4.02/student	₹41,276 ₹4.13/student	over budget
103.00 stress	₹12,311 ₹4.10/student	₹12,889 ₹4.30/student	₹41,357 ₹4.14/student	₹42,514 ₹4.25/student	over budget

The ceiling breaks at **₹95.89/USD** for Tier A and **₹96.91/USD** for Tier B on the On-Demand path. Both are roughly 2% above today's rate, which is thin. The mitigation is the quota dial: reducing the graded allowance by one evaluation per student per month recovers about ₹509 at Tier A and ₹1,696 at Tier B.

12.2 What could break the budget

Risk	Exposure	Mitigation
DeepSeek reasoning mode left enabled	+$256.69 to +$1,069.54/mo at Tier B	Set it off explicitly in the request; assert on the returned token counts in a test; alarm on output-tokens-per-call exceeding 700.
Rupee weakens past the breakeven rate	Breach at ₹96.91/USD	Reduce the graded quota by one per student per month; each step recovers roughly 1.5% of the bill. Review quarterly against the actual rate.
Spot capacity unavailable in ap-south-1	+$5.62 (A) / +$11.23 (B) per month	Already absorbed — every headline figure in this document is the On-Demand fallback.
Static assets stop being edge-cacheable	Up to +$134.66/mo at Tier B	A cache-busting or Cache-Control regression pushes all egress through AWS at $0.1093/GB. Alarm on the AWS DataTransfer-Out metric, not just on the Cloudflare hit ratio.
Concurrency exceeds the modelled peak	Compute only, and it is elastic	Target-tracking autoscaling is free and adds nodes on CPU. At Tier A the peak fleet saturates near 6,000 concurrent — 3.0× the stated peak.
The 40 req/s per vCPU assumption is optimistic	Needs more baseline vCPU	This is exactly what phase 5 measures before any reservation is bought. At half the assumed throughput Tier A still holds 1.5×.
Self-managed PostgreSQL fails and nobody has rehearsed a restore	Data loss; unbounded downtime	This is the single largest operational risk created by the budget. Phase 7 requires a timed restore rehearsal, not just a configured backup.
Graviton port blocked by an unexpected x86 dependency	+10–20% of compute	torch is the known one. Audit the full dependency tree for aarch64 wheels in phase 4, before reserving anything.

13 Verified rates and sources

Every AWS rate below was read this session from the AWS Price List Bulk API region file for `ap-south-1`, not from a third-party pricing site. The two third-party pages consulted during this work both returned us-east-1 figures while displaying a Mumbai region selector, which is how the 43% RDS premium came to be missed in earlier revisions.

Item	Basis	Rate	Source
EC2 c7g.xlarge (4 vCPU / 8 GiB)	On-Demand	`$0.0982 /hr`	Price List API
EC2 c7g.xlarge	3yr No-Upfront Standard RI	`$0.0445 /hr`	Price List API
EC2 c7g.large (2 vCPU / 4 GiB)	On-Demand	`$0.0491 /hr`	Price List API
EC2 m7g.large (2 vCPU / 8 GiB)	3yr No-Upfront Standard RI	`$0.0265 /hr`	Price List API
EC2 m7g.xlarge (4 vCPU / 16 GiB)	3yr No-Upfront Standard RI	`$0.0530 /hr`	Price List API
EBS gp3	provisioned storage	`$0.0912 /GB-mo`	Price List API
RDS PostgreSQL db.m7g.large	Single-AZ On-Demand	`$0.240 /hr`	Price List API
RDS PostgreSQL db.m7g.large	Single-AZ 1yr No-Upfront RI	`$0.1844 /hr`	Price List API
RDS gp3 storage	Single-AZ	`$0.131 /GB-mo`	Price List API
ElastiCache Valkey cache.t4g.micro	3yr No-Upfront RI	`$0.0080 /hr`	Price List API
ElastiCache Valkey cache.t4g.small	3yr No-Upfront RI	`$0.0168 /hr`	Price List API
Application Load Balancer	per ALB-hour	`$0.0239 /hr`	Price List API
Lambda arm64 duration	first tier	`$1.33334e-05 /GB-s`	Price List API
S3 Standard / Standard-IA	storage	`$0.025 / $0.0138 /GB-mo`	Price List API
Data transfer out (Mumbai to internet)	first 10 TB	`$0.1093 /GB`	Price List API
CloudWatch Logs, Infrequent Access	ingest	`$0.335 /GB`	Price List API
Groq whisper-large-v3-turbo	audio transcribed	`$0.04 /hr`	Groq pricing
Groq openai/gpt-oss-20b	input / output	`$0.075 / $0.30 per M tok`	Groq pricing
DeepSeek V3.2 (Bedrock Mantle)	input / output	`$0.62 / $1.85 per M tok`	AWS Bedrock
DeepSeek V3.2 cached input read	prompt cache hit	`$0.062 per M tok`	AWS Bedrock
USD / INR	Fed H.10, 4-Sep-2026	`Rs 94.49`	federalreserve.gov
EC2 Graviton Spot	assumed discount	`-65% vs On-Demand (NOT price-list verified)`	estimate

Price List API endpoint: `https://pricing.us-east-1.amazonaws.com/offers/v1.0/aws/<service>/current/ap-south-1/index.csv` for AmazonEC2, AmazonRDS, AmazonElastiCache, AWSLambda, AmazonS3, AWSELB, AmazonCloudWatch, AWSSecretsManager and AWSDataTransfer. **Exchange rate:** Federal Reserve H.10, release of 8 September 2026, rate for 4 September 2026. **Groq and Bedrock rates** are vendor-published list prices. **The Spot discount is the one estimate in this document** — Spot prices are not in the Price List API — and no headline figure depends on it.

13.1 Reservation structures available in Mumbai

One structural asymmetry is worth recording, because it shaped the design: **EC2 offers 3-year No-Upfront Standard Reserved Instances, and RDS does not.** RDS 3-year terms are All-Upfront or Partial-Upfront only, so the cheapest managed-database option also demands a large cash payment. EC2 gives 55–58% off with nothing up front. ElastiCache offers 3-year No-Upfront too. That is a second, independent reason the database sits on EC2.

Service	On-Demand	1yr No-Upfront	3yr No-Upfront	Discount at 3yr
EC2 c7g.xlarge	$71.69/mo	—	**$32.48/mo**	55%
EC2 m7g.large	$42.56/mo	—	**$19.34/mo**	55%
EC2 m7g.xlarge	$85.12/mo	—	**$38.69/mo**	55%
ElastiCache Valkey cache.t4g.small	$23.94/mo	—	**$12.26/mo**	49%
RDS PostgreSQL db.m7g.large (Single-AZ)	$175.20/mo	$134.61/mo	**not offered**	3yr is All- or Partial-Upfront only

14 Summary

	Tier A	Tier B
Registered students	3,000	10,000
Peak concurrent users	2,000	3,000
Daily active users	2,850	7,000
Budget ceiling per month	₹12,000	₹40,000
Infrastructure	$78.38	$198.85
AI and inference	$46.76	$213.91
Total per month	**$125.14**	**$412.76**
Total per month in rupees	**₹11,888**	**₹39,212**
Per registered student per month (worst case)	**₹3.96**	**₹3.92**
Headroom against the ceiling (worst case)	₹112	₹788
Expected bill under uneven usage (§2.1)	**₹9,788**	**₹29,092**
Expected per student per month	**₹3.26**	**₹2.91**
Expected headroom against the ceiling	₹2,212	₹10,908
Annual run rate	₹142,657	₹470,546
Peak serving headroom	3.0×	4.0×
Saturates at approximately	6,000 concurrent	12,000 concurrent
Graded evaluations per student per month	6	8
Practice turns per student per month	25	40
Database high availability	None — PITR restore	Streaming standby, manual promotion

Both tiers fit inside ₹4.16 per student per month on the pessimistic costing — Tier A by ₹112 and Tier B by ₹788 against the rounded ceilings, and by ₹592 and ₹2,388 against the arithmetic 4.16 limit. The infrastructure is straightforward. What the budget really costs is the twelve application changes in §8 — particularly the 5.1× reduction in cost per graded evaluation and the server-side quota — and the operational burden of running PostgreSQL yourself without automatic failover. Higher latency is not the price being paid here; the price is thinner redundancy, thinner observability, and less generous AI.